AI4REASON: Artificial Intelligence for Large-Scale Computer-Assisted Reasoning

Josef Urban

Czech Technical University in Prague
Czech Institute for Informatics, Robotics and Cybernetics
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Can You (or Your Machine) Prove the Pythagoras Theorem?

- What are those nerds on 6th floor doing all nights there?
- Formal statement and 15-line human-written verified proof:

```
:: The Pythagorean theorem
theorem :: BHSP_5:6
for X being RealUnitarySpace
for x, y being Point of X st x,y are_orthogonal holds
||.(x + y).|| ^2 = (||.x.|| ^2) + (||.y.|| ^2)
proof
let X be RealUnitarySpace; :: thesis:
let x, y be Point of X; :: thesis:
assume x,y are_orthogonal ; :: thesis:
then A1: x .|. y = 0 by BHSP_1:def 3;
A2: (x + y) .|. (x + y) >= 0 by BHSP_1:def 2;
A3: x .|. x >= 0 by BHSP_1:def 2;
A4: y .|. y >= 0 by BHSP_1:def 2;
||.(x + y).|| ^2 = (sqrt ((x + y) .|. (x + y))) ^2 by BHSP_1:def 4
.= (x + y) .|. (x + y) by A2, SQUARE_1:def 2
.= ((x .|. x) + (2 * (x .|. y))) + (y .|. y) by BHSP_1:16
.= ((sqrt (x .|. x)) ^2) + (y .|. y) by A1, A3, SQUARE_1:def 2
.= ((sqrt (x .|. x)) ^2) + ((sqrt (y .|. y)) ^2) by A4, SQUARE_1:def 2
.= (||.x.|| ^2) + ((sqrt (y .|. y)) ^2) by BHSP_1:def 4
.= (||.x.|| ^2) + (||.y.|| ^2) by BHSP_1:def 4 ;
hence ||.(x + y).|| ^2 = (||.x.|| ^2) + (||.y.|| ^2) ; :: thesis:
end;
```

Can You Prove the Pythagoras Theorem? - Details

- Our automatically found proof - the ENIGMA system: http://grid01.ciirc.cvut.cz/~mptp/enigma_prf/t6_bhsp_5.out
- http://grid01.ciirc.cvut.cz/~mptp/7.13.01_4.181.1147/html/bhsp_5.html#T6

```
# Proof object clause steps                : 181
# Proof object initial clauses used        : 51
# Proof object initial formulas used       : 34
# Proof object simplifying inferences      : 210
# Parsed axioms                            : 342
# Initial clauses in saturation            : 440
# Processed clauses                        : 8804
# ...remaining for further processing      : 4119
# Generated clauses                        : 65094
# ...of the previous two non-trivial       : 60807
# User time                  : 28.629 s
```

Can You Prove the Pythagoras Theorem? - Details

- Jan Jakubuv (the main developer of ENIGMA) has proved it automatically on May 6 2020 in 30s (are you that fast?)
- ENIGMA was guided by 150 decision trees each of max depth 80 and having maximum 8000 leaves
- Trained by the LightGBM gradient boosted tree toolkit (state of the art ML, fast).
- On a corpus of ca 500k previous automatically found proofs.
- We could not prove the theorem automatically before that even with much higher time limits.

Can You Prove the Pythagoras Theorem? - Details

- The proof attempt started with 342 mathematical facts.
- You can think of the facts as flat language sentences.
- In fact they are first-order logic parse trees that can interact in many ways.
- Preselected from a knowledge base (math library) of ca 150k math facts
- The initial facts logically interact, producing more and more facts (inferences)
- Their number quickly explodes (millions) unless some control is introduced
- In 30s this proof attempt has generated ”only” 60k more nontrivial facts

Can You Prove the Pythagoras Theorem? - Details

- Each of the 60k facts was scored by the 150 LightGBM decision trees
- I.e. on average we scored 2k facts per second (on a single commodity CPU)
- This is also thanks to our fast (but accurate and memory efficient) characterization of the facts by syntactic features
- The scoring was gradually choosing the best of the generated facts
- These are the ones used to perform inferences with the previously chosen (processed) ones
- This is called the given clause loop and it is the basis of today's strongest theorem provers

Can You Prove the Pythagoras Theorem? - Details

- ENIGMA gradually chose 4k facts from the 60k and did all possible inferences (modus-ponens style) among them
- When the last one was chosen, it interacted with the previous facts in such a way that the proof was finished
- In the end, only 34 of the 342 initial facts were needed for the proof
- And the proof needed only 181 steps, not 4k (and could generate much fewer than 60k facts)
- So if we were smarter, we could do it even faster!
- We could learn from this proof - saying which facts/inferences were good/bad
- And in the next proving attempt, we would probably do the proof better
- So we could interleave proving and learning from proofs
- And thus train better and better automated mathematicians!

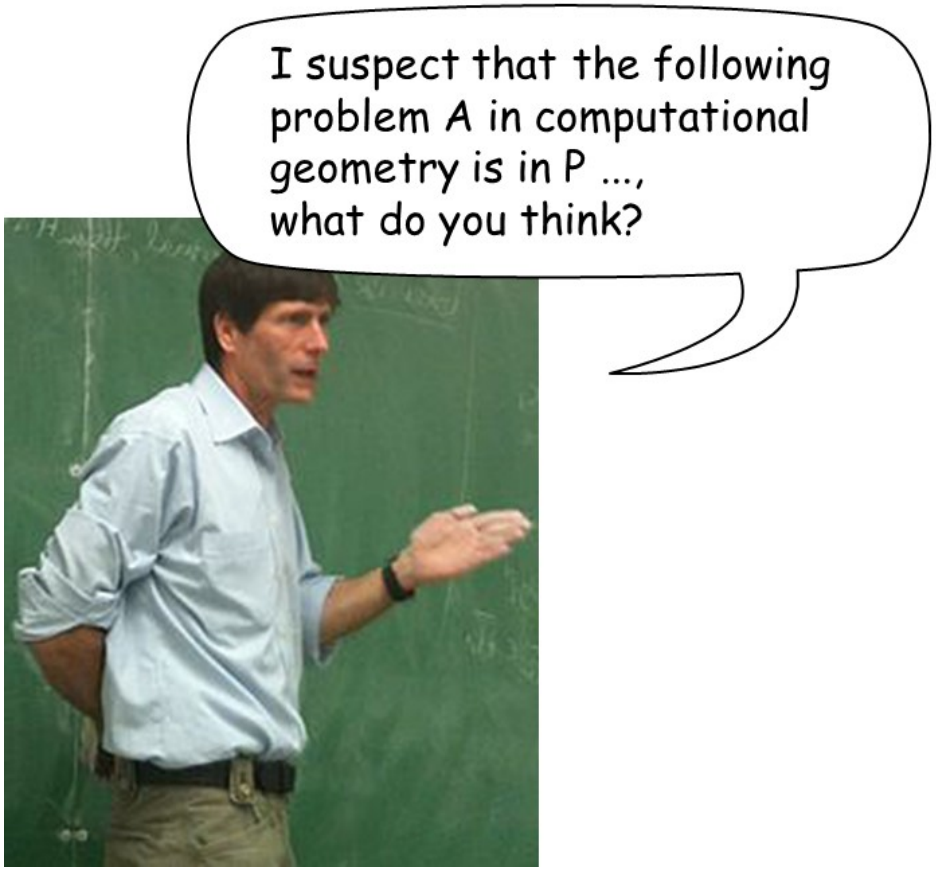
Feedback loop for ENIGMA on Mizar data

- Interleave proving and learning of ENIGMA guidance
- Done on 57880 Mizar problems very recently
- Ultimately a 70% improvement over the original strategy
- From 14933 proofs to 25397 proofs (all 10s - no cheating)
- As of 2021 we have 42519 proofs

	$\mathcal{S}$	$\mathcal{S} \odot \mathcal{M}_9^0$	$\mathcal{S} \oplus \mathcal{M}_9^0$	$\mathcal{S} \odot \mathcal{M}_9^1$	$\mathcal{S} \oplus \mathcal{M}_9^1$	$\mathcal{S} \odot \mathcal{M}_9^2$	$\mathcal{S} \oplus \mathcal{M}_9^2$	$\mathcal{S} \odot \mathcal{M}_9^3$
solved	14933	16574	20366	21564	22839	22413	23467	22910
$\mathcal{S}\%$	+0%	+10.5%	+35.8%	+43.8%	+52.3%	+49.4%	+56.5%	+52.8%
$\mathcal{S}+$	+0	+4364	+6215	+7774	+8414	+8407	+8964	+8822
$\mathcal{S}-$	-0	-2723	-782	-1143	-508	-927	-430	-845

	$\mathcal{S} \odot \mathcal{M}_{12}^3$	$\mathcal{S} \oplus \mathcal{M}_{12}^3$	$\mathcal{S} \odot \mathcal{M}_{16}^3$	$\mathcal{S} \oplus \mathcal{M}_{16}^3$
solved	24159	24701	25100	25397
$\mathcal{S}\%$	+61.1%	+64.8%	+68.0%	+70.0%
$\mathcal{S}+$	+9761	+10063	+10476	+10647
$\mathcal{S}-$	-535	-295	-309	-183

Not So Distant Future



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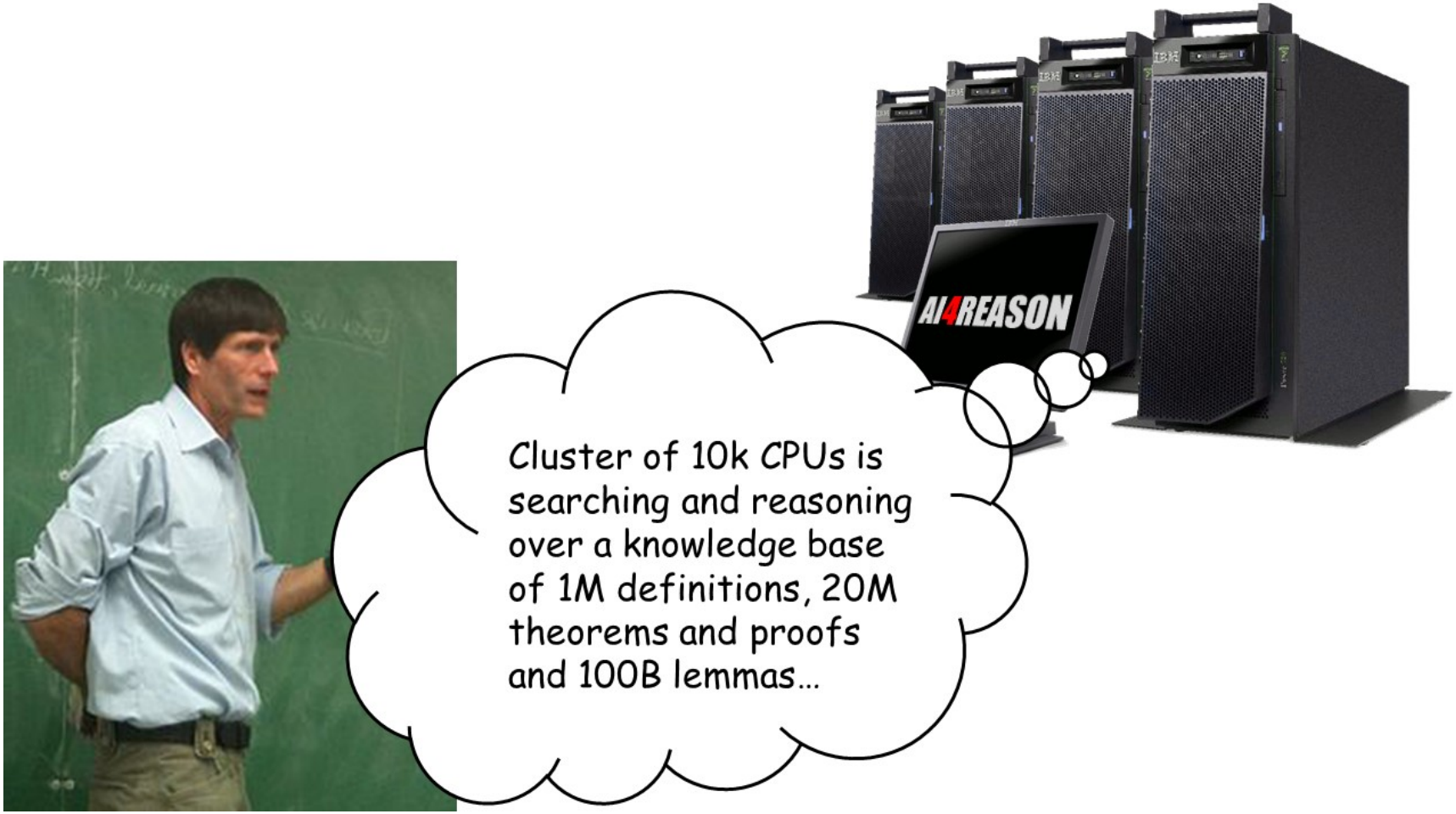

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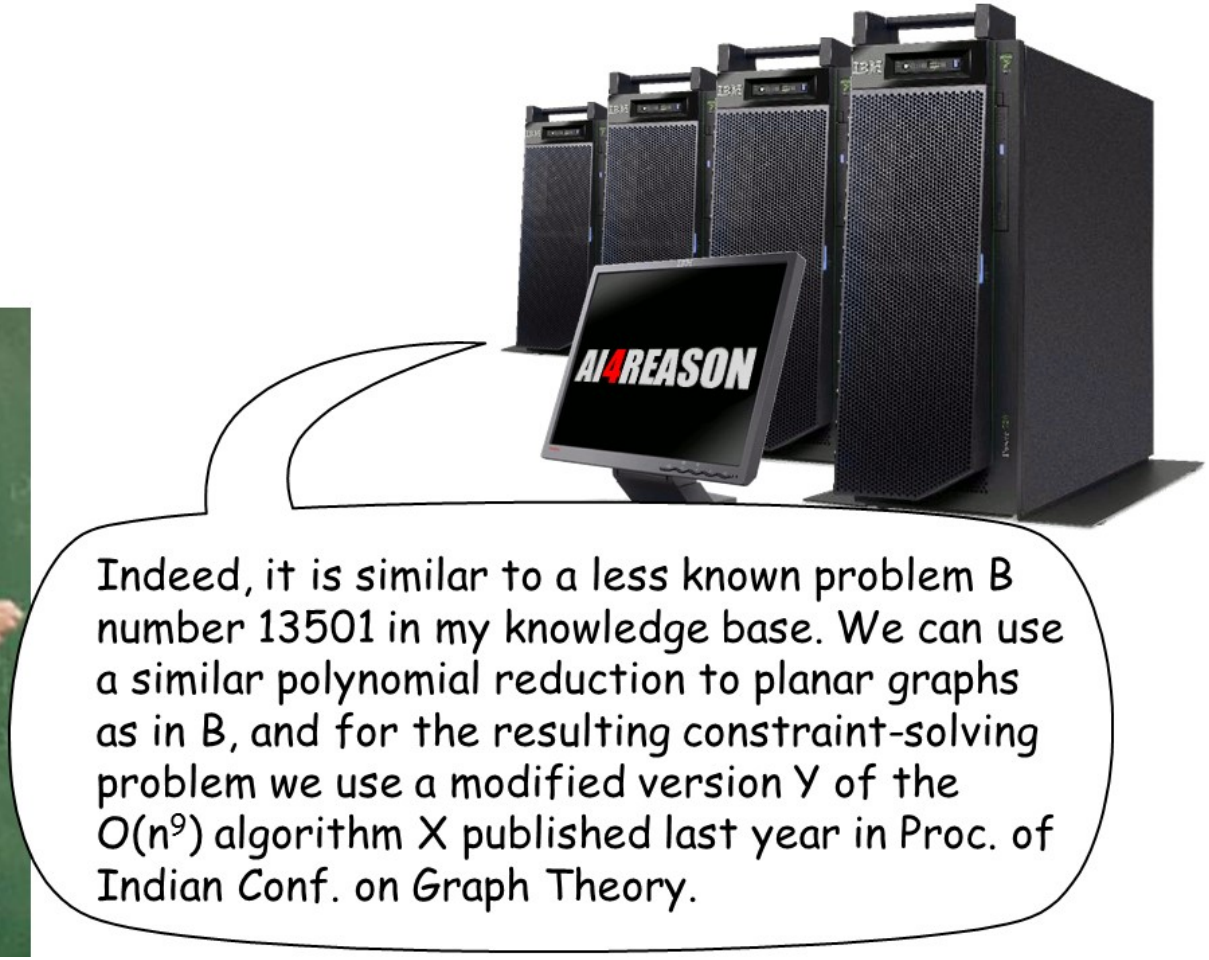

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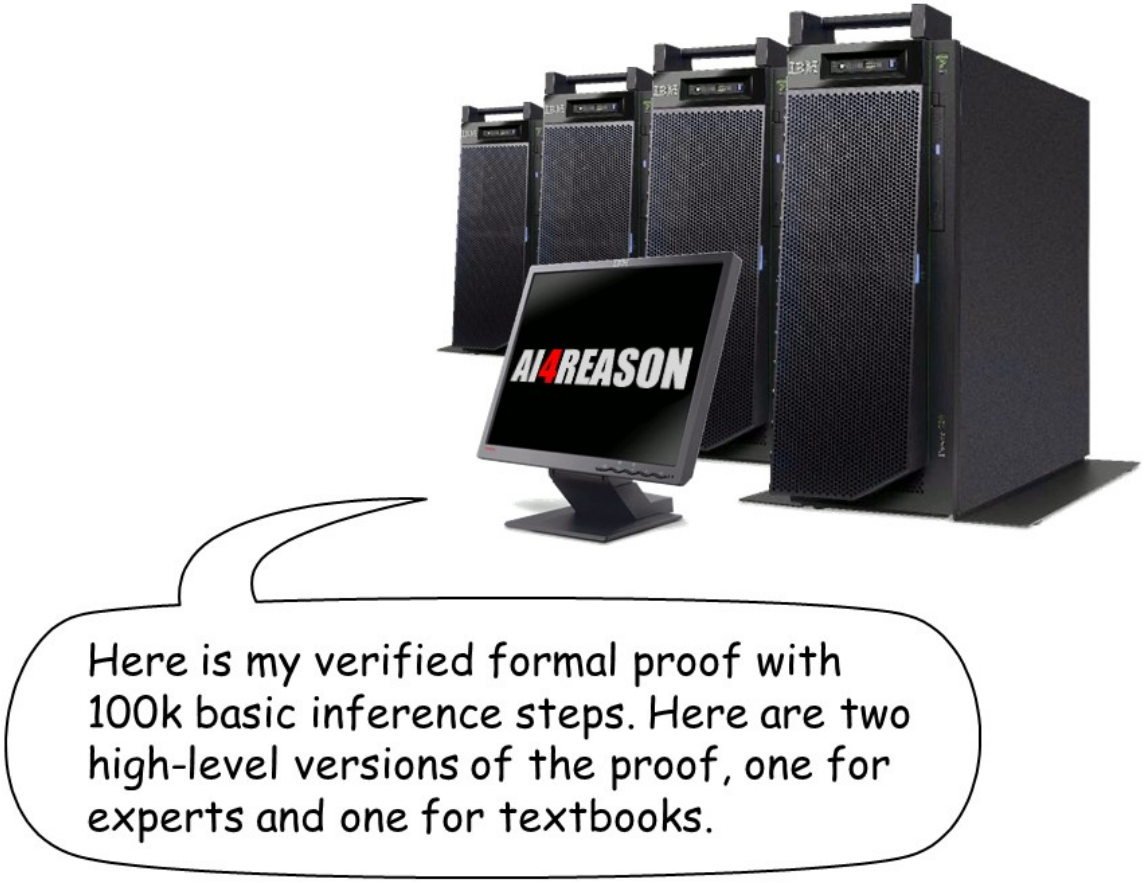


Not So Distant Future

Btw., A, B and X, Y generalize to a far-reaching conjecture that could solve a long-standing open problem.

Let's write an ERC proposal about exploring them!

How Distant?

- 15 - 50 years, depending on our efforts
- Today's numbers about 100x smaller:
  - 10k-30k computer-understandable definitions
  - 200k-300k (small) theorems and proofs
  - 1B-10B primitive lemmas
- Covers roughly the Bc level in Math/CS, PhD level still far
- The main bottleneck:

WEAK AUTOMATION OF REASONING
OVER LARGE COMPUTER-UNDERSTANDABLE CORPORA

- This is where a breakthrough is necessary

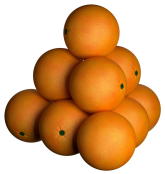
AI4REASON Goals

- Breakthrough in a hard problem in AI and reasoning: automatically proving theorems in complex theories
- Produce AI systems that combine learning and reasoning
- Thus help with automating verification of:
  - advanced mathematics and big proofs (Kepler conjecture)
  - software and hardware designs (seL4 OS microkernel)
  - advanced systems and designs (finance, industry, science)
- The idealized/perfect World of Math (Plato/Gödel): Interesting AI area – narrow or general AI?

Example: The Kepler conjecture

- J. Kepler (1611, Prague): The most compact way of stacking balls of the same size in space is a pyramid.

$$V = \frac{\pi}{\sqrt{18}} \approx 74\%$$



- Big proof: 300 pages + computations (Hales, Fergusson, 1998)
- Formal proof finished in 2014, 20000 theorems & proofs
- All of it computer-understandable and verified in HOL Light:
- `polyhedron s /\ c face_of s ==> polyhedron c`
- However, this took 20 – 30 person-years!
- Our AI methods can fully automate 40% of the proofs (2014)
- Similar verification efforts for bug-free compilers, OS, etc.

Sample of Formal Math: Irrationality of $\sqrt{2}$

```
theorem sqrt2_not_rational:
  "sqrt (real 2) ∉ ℚ"
proof
  assume "sqrt (real 2) ∈ ℚ"
  then obtain m n :: nat where
    n_nonzero: "n ≠ 0" and sqrt_rat: "¦sqrt (real 2)¦ = real m / real n"
    and lowest_terms: "gcd m n = 1" ..
  from n_nonzero and sqrt_rat have "real m = ¦sqrt (real 2)¦ * real n" by simp
  then have "real (m²) = (sqrt (real 2))² * real (n²)"
    by (auto simp add: power2_eq_square)
  also have "(sqrt (real 2))² = real 2" by simp
  also have "... * real (m²) = real (2 * n²)" by simp
  finally have eq: "m² = 2 * n²" ..
  hence "2 dvd m²" ..
  with two_is_prime have dvd_m: "2 dvd m" by (rule prime_dvd_power_two)
  then obtain k where "m = 2 * k" ..
  with eq have "2 * n² = 2² * k²" by (auto simp add: power2_eq_square mult_ac)
  hence "n² = 2 * k²" by simp
  hence "2 dvd n²" ..
  with two_is_prime have "2 dvd n" by (rule prime_dvd_power_two)
  with dvd_m have "2 dvd gcd m n" by (rule gcd_greatest)
  with lowest_terms have "2 dvd 1" by simp
  thus False by arith
qed
```

```
let SQRT_2_IRRATIONAL = prove
 (`~rational(sqrt(&2))`,
  SIMP_TAC[rational; real_abs; SQRT_POS_LE; REAL_POS] THEN
  REWRITE_TAC[NOT_EXISTS_THM] THEN REPEAT GEN_TAC THEN
  DISCH_THEN(CONJUNCTS_THEN2 ASSUME_TAC MP_TAC) THEN
  SUBGOAL_THEN `~((&p / &q) pow 2 = sqrt(&2) pow 2)`
    (fun th -> MESON_TAC[th]) THEN
  SIMP_TAC[SQRT_POW_2; REAL_POS; REAL_POW_DIV] THEN
  ASM_SIMP_TAC[REAL_EQ_LDIV_EQ; REAL_OF_NUM_LT; REAL_POW_LT;
               ARITH_RULE `0 < q <=> ~(q = 0)`] THEN
  ASM_MESON_TAC[NSQRT_2; REAL_OF_NUM_POW;
                REAL_OF_NUM_MUL; REAL_OF_NUM_EQ]);;
```

The AI4REASON Plan of Attack

WP1 AI for finding relevant knowledge in large formal corpora:

- How to capture similarity and analogy of ideas?
- How to learn from proofs, counter-examples and theories?

WP2 AI-based guiding methods for reasoning tools:

- How to efficiently apply the learned guidance?
- How to automatically learn the best reasoning strategies?

WP3 AI for suggesting plausible conjectures and concepts:

- What makes a good conjecture for a given problem?
- What concepts are good for a given problem?

WP4 Self-improving AI interleaving learning and deduction:

- How to explore easier problems to learn for harder ones?
- How to develop theories and gain most useful knowledge?

WP5 Deployment and Cross-Corpora Reuse:

- Deploy the methods as strong online services
- Translate informal math to formal

Combining Learning and Theorem Proving

- **high-level**: select relevant lemmas from a large library
- **high-level**: select good high-level strategies for a problem
- **low-level**: guide all inference steps of theorem provers
- **mid-level**: guide application of tactics to goals
- **mid-level**: invent suitable strategies for problem classes
- **mid-level**: invent suitable conjectures for a problem
- **mid-level**: invent suitable concepts/models for problems
- **proof sketches**: explore related theories to get proof ideas
- **theory exploration**: develop new theories by conjecturing
- **feedback loops**: (dis)prove, learn from it, (dis)prove more
- ...

Some Highlights

- Won several divisions of the 2015-20 proving competitions (CASC)
- 2017-21: Improved the best open prover by ML guidance (ENIGMA - large breakthrough in 2019)
- 2018: 40% improvement of the leanCoP prover by reinforcement learning
- 2017-18: TacticToe – first ML-guided tactical system
- 2015-18: Blind Strategymaker - invent proving strategies
- First deep-learning based provers (with Google Research)
- 2015-18: Inf2formal – Translating informal math to formal, using grammar-based/semantic and neural systems
- Invited talks – Fields Inst., TYPES'18, Hales'60, AGI'18
- 2017 Google Research Award for JU
- AITP conference series started: `aitp-conference.org`
- AI/TP groups at Google Research (2016), OpenAI, Facebook, Microsoft, Amazon

What Can We Automatically Prove?

Nontrivial human-written proof that face of a polyhedron is polyhedron:

```
let FACE_OF_POLYHEDRON_POLYHEDRON = prove
 (`!s:real^N->bool c. polyhedron s /\ c face_of s ==> polyhedron c`,
  REPEAT STRIP_TAC THEN FIRST_ASSUM
   (MP_TAC o GEN_REWRITE_RULE I [POLYHEDRON_INTER_AFFINE_MINIMAL]) THEN
  REWRITE_TAC[RIGHT_IMP_EXISTS_THM; SKOLEM_THM] THEN
  SIMP_TAC[LEFT_IMP_EXISTS_THM; RIGHT_AND_EXISTS_THM; LEFT_AND_EXISTS_THM] THEN
  MAP_EVERY X_GEN_TAC
   [`f:(real^N->bool)->bool`; `a:(real^N->bool)->real^N`;
    `b:(real^N->bool)->real`] THEN
  STRIP_TAC THEN
  MP_TAC(ISPECL [`s:real^N->bool`; `f:(real^N->bool)->bool`;
                 `a:(real^N->bool)->real^N`; `b:(real^N->bool)->real`]
         FACE_OF_POLYHEDRON_EXPLICIT) THEN
  ANTS_TAC THENL [ASM_REWRITE_TAC[] THEN ASM_MESON_TAC[]; ALL_TAC] THEN
  DISCH_THEN(MP_TAC o SPEC `c:real^N->bool`) THEN ASM_REWRITE_TAC[] THEN
  ASM_CASES_TAC `c:real^N->bool = {}` THEN
  ASM_REWRITE_TAC[POLYHEDRON_EMPTY] THEN
  ASM_CASES_TAC `c:real^N->bool = s` THEN ASM_REWRITE_TAC[] THEN
  DISCH_THEN SUBST1_TAC THEN MATCH_MP_TAC POLYHEDRON_INTERS THEN
  REWRITE_TAC[FORALL_IN_GSPEC] THEN
  ONCE_REWRITE_TAC[SIMPLE_IMAGE_GEN] THEN
  ASM_SIMP_TAC[FINITE_IMAGE; FINITE_RESTRICT] THEN
  REPEAT STRIP_TAC THEN REWRITE_TAC[IMAGE_ID] THEN
  MATCH_MP_TAC POLYHEDRON_INTER THEN
  ASM_REWRITE_TAC[POLYHEDRON_HYPERPLANE]);;
```

We find an alternative shorter proof based on learning from the large library.

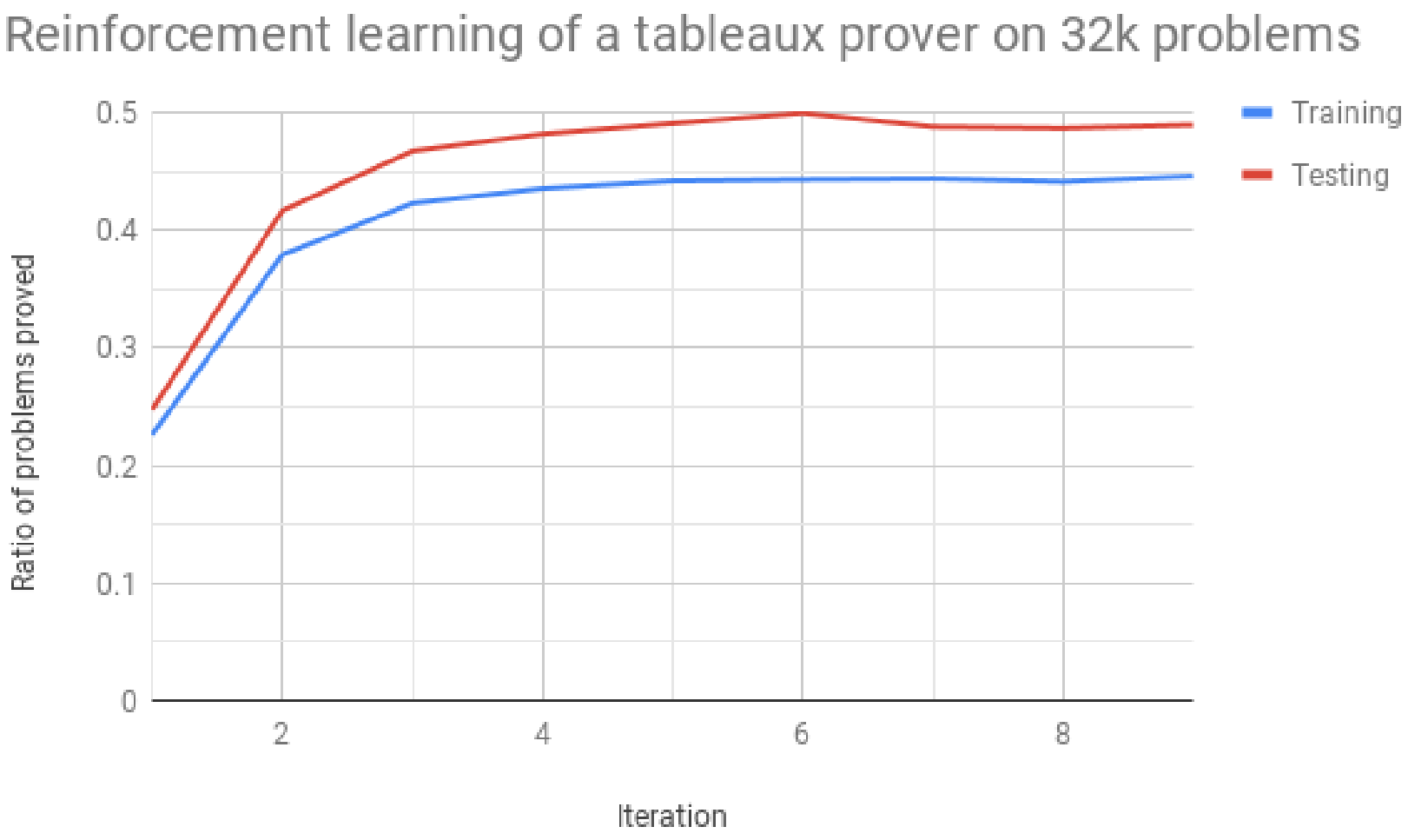
Statistical/Symbolic Guidance by Related Proofs (ProofWatch)

```
theorem Th36: :: YELLOW_5:36
for L being non empty Boolean RelStr for a, b being Element of L
holds ( 'not' (a "\/" b) = ('not' a) "/\" ('not' b)
         & 'not' (a "/\" b) = ('not' a) "\/" ('not' b) )
```

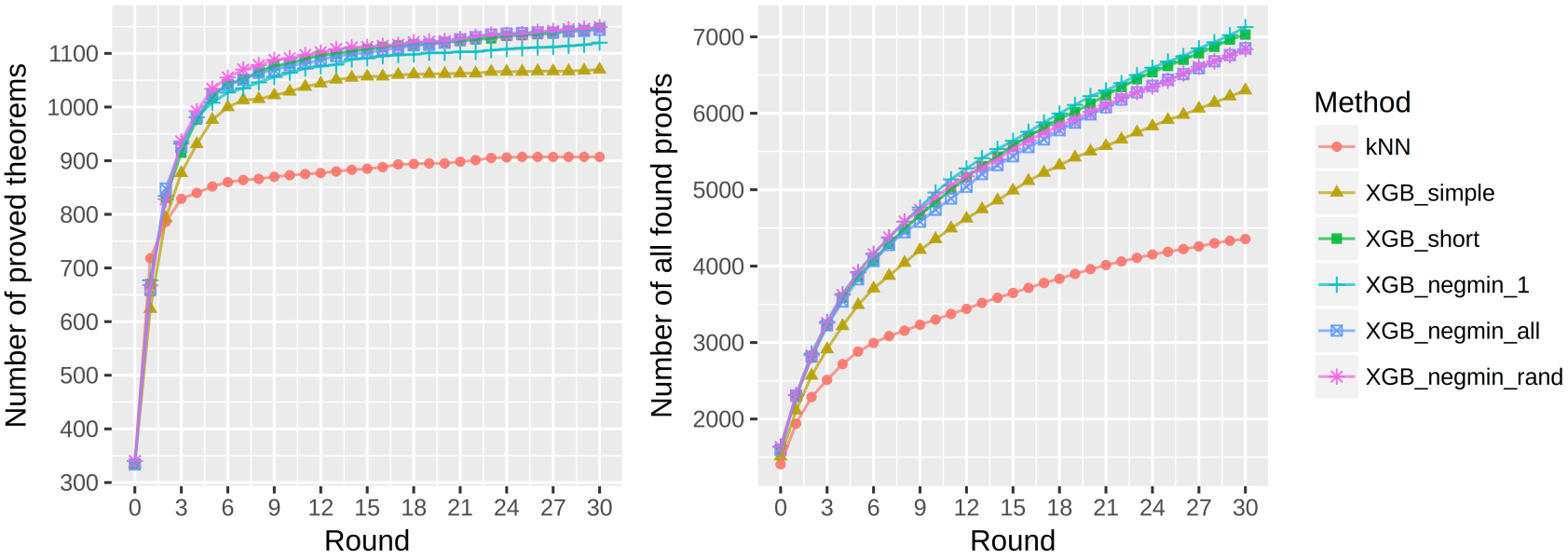
- Nontrivial proof of De Morgan laws for Boolean lattices
- Guided by continuous matching against 32 related proofs
- Most helped by a proof of a related statement for lower-bounded Heyting algebras

```
theorem :: WAYBEL_1:85
for H being non empty lower-bounded RelStr st H is Heyting holds
for a, b being Element of H holds
  'not' (a "/\" b) >= ('not' a) "\/" ('not' b)
```

Reinforcement Learning of a Tableaux Prover



Feedback Loop Generating Alternative Proofs



Informal2formal: Statistical/Semantic Parsing of Math

Strong-semantics probabilistic parser for HOL Light

Input the formula to parse. Separate symbols with spaces:

sin 0 = cos pi / 2 [Submit]

debug: cache→ decode → 18 bigram&trigram features → 1024 nearest neighbours → 16 nyk parses → 12 distinct terms

Conjecture as HOL Light term:	Type info:	Automatically Provable?		Time
sin (&0) = cos pi / &2		disproved		(6.74s)
sin (&0) = cos (pi / &2)		yes	REWRITE_TAC [SIN_0; COS_PI2]	(0.87s)
csin (Cx (&0)) = Cx (cos (pi / &2))		yes	REWRITE_TAC [CSIN_0; COS_PI2]	(0.74s)
csin (Cx (&0)) = ccos (Cx (pi / &2))		yes	MESON_TAC [NUMERAL; CX_COS; CSIN_0; COS_PI2]	(0.76s)
Cx (sin (&0)) = ccos (Cx (pi / &2))		yes	MESON_TAC [SIN_0; NUMERAL; CX_COS; COS_PI2]	(0.70s)
Cx (sin (&0)) = Cx (cos (pi / &2))		yes	REWRITE_TAC [SIN_0; COS_PI2]	(0.80s)

exp (ii * x) = ii * (sin x) + (cos x)

Conjecture as HOL Light term:	Type info:	Automatically Provable?	
cexp (ii * A0) = ii * (csin A0 + ccos A0)	A0:real^2	no advice	
cexp (ii * A0) = ii * csin A0 + ccos A0	A0:real^2	yes	REWRITE_TAC [COMPLEX_ADD_SYM; CEXP_EULER]
cexp (ii * Cx A0) = ii * Cx (sin A0 + cos A0)	A0:real	disproved	
cexp (ii * Cx A0) = ii * (csin (Cx A0) + ccos (Cx A0))	A0:real	no advice	
cexp (ii * Cx A0) = ii * csin (Cx A0) + ccos (Cx A0)	A0:real	yes	REWRITE_TAC [COMPLEX_ADD_SYM; CEXP_EULER]

Neural Informal2formal: Performance after Training

Rendered LaTeX	Suppose s_8 is convergent and s_7 is convergent . Then $\lim(s_8+s_7) = \lim s_8 + \lim s_7$
Input LaTeX	`Suppose $ { s _ { 8 } } $ is convergent and $ { s _ { 7 } } $ is convergent . Then $ \mathop { \rm lim } ( { s _ { 8 } } { + } { s _ { 7 } } ) \mathrel { = } \mathop { \rm lim } { s _ { 8 } } { + } \mathop { \rm lim } { s _ { 7 } } $ .`
Correct	`seq1 is convergent & seq2 is convergent implies lim ( seq1 + seq2 ) = ( lim seq1 ) + ( lim seq2 ) ;`
Snapshot-1000	`x in dom f implies ( x * y ) * ( f | ( x | ( y | ( y | y ) ) ) ) = ( x | ( y | ( y | ( y | y ) ) ) ) ) ;`
Snapshot-2000	`seq is summable implies seq is summable ;`
Snapshot-3000	`seq is convergent & lim seq = 0c implies seq = seq ;`
Snapshot-4000	`seq is convergent & lim seq = lim seq implies seq1 + seq2 is convergent ;`
Snapshot-5000	`seq1 is convergent & lim seq2 = lim seq2 implies lim_inf seq1 = lim_inf seq2 ;`
Snapshot-6000	`seq is convergent & lim seq = lim seq implies seq1 + seq2 is convergent ;`
Snapshot-7000	`seq is convergent & seq9 is convergent implies lim ( seq + seq9 ) = ( lim seq ) + ( lim seq9 ) ;`
Snapshot-12000	`seq1 is convergent & seq2 is convergent implies lim ( seq1 + seq2 ) = ( lim seq1 ) + ( lim seq2 ) ;`

Team and Collaborations

- Chad Brown, Jan Jakubův, Martin Suda, Thibault Gauthier, Bartosz Piotrowski, Zarathustra Goertzel, Shawn Wang
- plus several students and researchers on OPVVV funding (Janota, Chvalovsky, Nagashima, Blaauwbroek)
- now their own grants: Janota - ERC-CZ, Suda - GACR
- External scientific advisors
  - Prof. Stephan Schulz (Autom. reasoning, DHBW Stuttgart)
  - Prof. Robert Veroff (Autom. reasoning, U. of New Mexico)
  - Prof. Tom Heskes (AI, Radboud U. Nijmegen)
- Further Collaborations
  - Dr. Cezary Kaliszyk, U. of Innsbruck (ERC in 2016)
  - Dr. Jasmin Blanchette, VU Amsterdam (ERC in 2016)
  - Prof. Larry Paulson, U. of Cambridge (ERC in 2017)
  - Prof. Geoff Sutcliffe, U. of Miami
  - Dr. Christian Szegedy, Google Research
  - Prof. Herman Geuvers, Radboud U. Nijmegen
- over 40 research visits so far
- large related national grant awarded to JU in 2017

Future Potential - Science

- Use strong AI/reasoning and formal verification for:
- Science
  - Routinely verify complex math, software, hardware?
  - Make all of math/science computer-understandable?
  - Strong AI assistants for math/science?
- Examples
  - Automatically understand/verify/explain all arXiv papers?
  - Can we train a superhuman system like AlphaGo/Zero for math/physics? What will it take?
  - Can we prove that the Amazon Cloud cannot be hacked?
  - The same for critical government/private IT systems?

Future Potential - Society

- Use strong AI/reasoning and formal verification for:
- Society
  - Leibniz's dream: Let us Calculate! (solve any dispute)
  - J. McCarthy: Mathem. Objectivity and the Power of Initiative
  - AI/reasoning assistants for law/regulations
  - Verification of financial, transport/traffic systems, ...
  - Explainable and very securely verified systems
- Examples
  - Prove that two Paris metro trains will never crash?
  - Prove that a trading system doesn't violate regulations?
  - Prove that a new law is inconsistent with an old one?
  - Automatically debunk fallacies in political campaigns?

Links and Impacts on Other AI Areas

- Main areas: Machine Learning, Automated Reasoning
- Needs advances in Representation Learning
- AI needs intuition, but also reasoning and explanations
- Impact on Formal Verification (SW, HW, etc.)
- Potentially on any (hard) science/thinking/arguing
- Alan Turing, 1950, AI:

"We may hope that machines will eventually compete with men in all purely intellectual fields."

Outlook - Bets from 2014

- In 20 years, 80% of Flyspeck and Mizar toplevel theorems will be provable automatically (about 40% in 2014)
- The same in 30 years - I'll give you 2:1, In 10 years: 60% (getting there)
- In 25 years, 50% of the toplevel statements in LaTeX-written Msc-level math curriculum textbooks will be parsed automatically and with correct formal semantics

Outlook – Scientific Revolution

- (from a talk about Kepler and Hales)
- What did Kepler, Galileo & Co start to do in 1600s?
- What are we trying to do today?
- Kepler's Conjecture in Strena in 1611 (with many others)
- Kepler's laws, Newton, ..., age of science, math, machines
- ..., Hilbert, ..., Turing, ... age of computing machines?
- 1998 machine helps to find a proof of Kepler's Conjecture
- 2014 machine verifies a proof of Kepler's Conjecture
- ... 2050? machine finds a proof of Kepler's Conjecture?
- (no betting ;-)

Experience in Czech republic

- The Good
- The Bad
- The Ugly

Institutional Support

- Great support from the CIIRC team headed by V. Mařík
- Competent administrators at CIIRC (comparable to NL)
- Active CIIRC support for new large projects
- Not all perfect, but people are seriously trying
- EURAXESS - support for work visa for non-EU foreigners
- In 2016 we employed Dr. Chad Brown (US, PhD from CMU) – world's top expert in higher-order theorem proving
- Four international PhD students in 2017 – US, JP, PL, NL - one finished now (but not in CZ!)
- Excellent young postdoc hired in 2018: Thibault Gauthier
- Great returning more senior AR people in 2018: Martin Suda, Mikolas Janota (CIIRC's tenure track)

Institutional Support

- More good things:
  - We have managed to employ Chad Brown – world's top expert in higher-order theorem proving
  - Many extended (1 month and more) visits of senior and junior researchers to our group
  - Many short visits and seminar talks
  - Today I am giving the first talk in a new seminar series introduce in our field at Harvard
  - We have brought the CLAIRE (and others the ELLIS) AI network to CZ
  - I am in CLAIRE's Extended Core Team, we have brought the CLAIRE office to Prague

My Experience in Czech republic – the Bad

- Bureaucracy – conference and research travel, rigid salary rules, strange internal rules about employement, public tender needed to buy a notebook!
- ERC should impose minimal salaries of PIs in each EU country (like in Marie Curie grants)
- Unbelievable "scientific" system of scientific evaluation: "Jachyme, hod ho do stroje"
- How did Thomson Reuters with its mediocre products manage to impose monopoly in this country??
- no Czech financial/tax support for attracting "knowledge workers" (compare with NL, DK, etc.)

My Experience in Czech republic – the Bad

- Minimal institutional financing - needs to change rapidly!
- Ministry of Education very bureaucratic in OPVVV grant management compared to ERC
- Incomparably higher amount of (nonsensical) regulations and reporting required by the Ministry compared to ERC
- The Ministry is quite incompetent in grant management and reviewing compared to not just ERC, but NWO and GACR
- Researchers are treated as criminals in a prison by the Ministry - absolute lack of trust and flexibility
- This goes against Ministry's declared goal to attract top researchers from abroad
- If this does not change, nobody good will stay here - there are great opportunities, money and flexibility in AI research around the world today

My Experience in Czech republic – the Bad

- Rigid and study-oriented PhD system in CZ (FEL!)
- Unsuitable for good international students - practically all my foreign students quit FEL by now
- Not competitive with research-oriented PhD in NL, AT
- All my new PhD students now start in NL or AT
- Appallingly incompetent leadership/management (rector) of CTU - I nearly left in 2018
- Bad system of choosing university management in CZ
- Democracy is not optimal, students should not govern
- Universities in the Netherlands are governed meritocratically and by experienced management teams
- There is a lot of feedback and control in the system and it works much better than in CZ
- The CZ university law critically needs to change and adopt the NL system!

Experience in Czech republic – the Ugly

- Why are we not speaking in English here today?
- Are there no foreigners working in Czech science?
- Can we get a bit more relaxed in science here?
- Friday happy hour with beer in NL/UK, sofas, table-tennis, lots of other perks for researchers at Google, ...
- If not, we will not be competitive in attracting the best students, postdocs, researchers ...

Thanks

- Thanks for your attention!
- Questions?